

NEXORA SYSTEM ARCHITECTURE MANUAL

Complete Analysis of the Live Full-Stack Student Hub Application

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Client: Nexora Development Team

Tech Stack: React Native, Expo Web, Supabase

Release State: Production Ready (v1.0)

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1. Executive Summary & Project Overview

Nexora is a modern full-stack academic and career launching pad application built to guide students through the discovery of vocational and professional pathways. By linking local educational structures with industrial trades and elite universities, Nexora offers a personalized hub for career development. The application leverages a hybrid online/offline dataset architecture, allowing seamless connectivity via Supabase Cloud, and is optimized for cross-platform deployment on Android, iOS, and Web targets.

The system has been engineered with a futuristic 'cyberpunk-neon' style design system, offering dark and light mode adaptivity, interactive animations, and robust verification steps designed to impress the modern student demographic. Key achievements include automated OAuth credential resolution, an interactive career selection matrix, and an automated match-scoring algorithm that matches college campus missions directly against user-curated skill dossiers.

2. Core Technology Stack & Architecture

The application leverages modern web and mobile standards to provide absolute flexibility, speed, and reliability. Below are the core technological layers and dependencies utilized in this project:

- **Frontend Framework:** React Native built on top of Expo SDK 54, enabling shared TypeScript codebases across web, Android, and iOS devices.
- **Web Deployment Engine:** React Native Web (react-native-web) integrating Vercel hosting configurations for high-speed instant preview and deployment.
- **Navigation Router:** React Navigation including Native Stack Router and Bottom Tab Navigation engines configured with custom animated interfaces.
- **Backend Database & Auth Service:** Supabase Platform (@supabase/supabase-js) providing secure database storage, User Profile Dossiers, Mock/Real College catalogs, and email/Google OAuth authentication services.
- **Aesthetics & Graphics:** Expo Linear Gradient backgrounds, Ionicons iconography packs, and customized React Native Animated library modules (spring animations, rotation tracers, and pulse overlays).

3. Student Onboarding & Navigation Flow

To ensure a structured and engaging user experience, the application utilizes a specific state-driven onboarding and routing lifecycle. Below is the step-by-step progression of how a student navigates from initial authorization to the main dashboard:

- **1. Session Verification:** On launch, App.tsx queries Supabase auth.getSession() and listens to onAuthStateChange(). If no session is found, the user is restricted to the AuthStack. On successful authentication, the authenticated Stack.Group is mounted, rendering the Sector Selection screen first.
- **2. Command Hub Entry:** SectorSelectionScreen.tsx plays a typewriter intro ('NEXORA' -> 'CHOOSE PATH'). The user is then presented with 14 interactive career pods. Selecting any sector other than 'ENGINEERING' triggers a direct navigation.replace('Home') routing them straight to the main Dashboard.
- **3. Guided Engineering Branching:** If the user selects 'ENGINEERING', the screen navigates to EngineeringPathScreen.tsx. The student is guided through a sequence of career choices (e.g. Intermediate, Diploma, ITI -> Science, Commerce, Arts -> MPC, Mechanical Engineering, Turner, etc.).
- **4. Pathway Synchronization & Dashboard Landing:** Once a student reaches a terminal option in the career tree, a status check displays 'Pathway Synchronized'. After a 2000ms delay, the router executes navigation.replace('Home'), landing the student on the bottom-tab navigation hub (TabNavigator) which defaults to the Home screen dashboard.

4. Screen-by-Screen Feature Specifications

The Nexora application has been fully modularized into discrete screens, each serving a core requirement in the student's journey. Below is the detailed specification of the implemented screens:

A. Authentication Gate (AuthScreen.tsx)

The security hub implements multiple security entry vectors. It provides traditional email and password authentication, password recovery hubs, security key updates, and quick One-Time Password (OTP) logins. Additionally, it integrates a Google OAuth gateway that handles redirects on the web and links back via deep linking (Linking.createURL) in Expo Go environments. The screen features a dynamic tracer animation that spins around the gateway card and a breathing brand logo animation.

B. Sector Selection Screen (SectorSelectionScreen.tsx)

This serves as the initial landing screen after authorization. It triggers a typewriter intro animation sequence ('NEXORA' and 'CHOOSE PATH') and renders a grid of 14 core industrial sectors (e.g., Engineering, Skilled Trades, Computers, Hospitality, fashion). Each item is represented as a pod with circular color borders, spring scaling effects upon user touch/hover, and neon-themed glow halos. Selecting Engineering redirects the user to the Career Tree, while other sectors navigate to the main Hub.

C. Career Decision Tree (EngineeringPathScreen.tsx)

This interactive module maps a student's educational background to potential industrial targets. It acts as a decision tree guiding students from foundation choices (Intermediate, Diploma, or ITI) through streams (Science, Commerce, Arts) to precise branches (Civil, Mechanical, Computer Engineering, Electrician, Machinist). The screen incorporates sliding card animations, a pulsing progress bar, and descriptive panels clarifying the nature and scope of each career path before final synchronization.

D. Student Hub Dashboard (HomeScreen.tsx)

The central hub contains the student dashboard. Its primary element is the animated Career Launchpad card which pulses dynamically to show the student's current matching percentage (e.g. 84% matched). The screen includes horizontal scrolling cards featuring scholarship applications (e.g. Global STEM, Google Pro), an interactive 4-tile grid (Scholarships, Resume Lab, Community, Career Quiz) with customized divider layouts, an exit gate (sign-out button linked to Supabase), and a calendar row showing upcoming events.

E. College Hub Screen (CollegeScreen.tsx)

This comprehensive university search and application module queries the Supabase database. It indicates live connectivity status with an online/offline cache sync badge. Students can search via text queries or filter by sector tabs. Each college card features a customized matching badge. Clicking on a college card displays the Mission Brief Modal with core directives, tuition waiver information, and specific entry requirements. The application gateway lets students apply directly, creating a unique Admissions Token (e.g. NEX-723049) and upserting the application to the database.

F. Student Profile Dossier (ProfileScreen.tsx)

A LinkedIn-style profile manager enabling users to edit their biographical details, educational matrix (high school and college trade details), contact information, and social links (LinkedIn URL). It features an interactive Skill Tag Manager allowing students to add/remove custom skill badges in real time. The Dossier is synchronized directly with the Supabase profiles database table. The updated skill portfolio dynamically feeds into the matching algorithms of the College Hub.

G. Secondary Modules (AIScreen.tsx, NotificationScreen.tsx)

- Nexora AI Screen: Houses the chatbot S-Node interface allowing users to record or type academic doubt queries.
- Notification Screen: Logs system notifications confirming database sync, profile status updates, and admissions activity.

5. Key Technical Highlights & Custom Algorithms

Beyond standard forms and screens, the Nexora application implements several specialized logical systems that enhance data integrity, user experience, and personalized intelligence:

I. Skill-to-College Match Rate Algorithm

Each college card displays a personalized match rate. If the user's Profile Dossier contains skills (e.g., Python, CAD), the College screen loops through the college's campus mission, Perks, and Description arrays. For each skill keyword found in these text descriptions, the baseline match rate is boosted by 3%, capped at a maximum of 100%. This provides students with immediate visual reinforcement about which institutes match their active skillset. If no custom skills are present, the algorithm reverts to the baseline institutional match rate.

II. Offline Cache & Supabase Network Resiliency

The application implements a robust hybrid data retrieval pattern. When fetching college information, the client first attempts to establish a connection with the remote Supabase 'colleges' table. If the database is unreachable or offline, the system catches the error, transitions the sync badge from 'ONLINE SYNCED' to 'CACHE SYNCED', and seamlessly loads a robust mock database from the offline client memory. This ensures zero service disruption for students operating in low-bandwidth environments.

III. Dynamic Animated Feedback Engine

To achieve a premium, futuristic aesthetic, several animations are ran concurrently: 1. Glowing Halo effects that calculate positioning dynamically to expand outwards when a card is focused. 2. Interleaved spring animations that control icon scaling, text slides, and opacity triggers in the tab navigation bar. 3. A constant rotating linear gradient trace border wrapping around the login gateway card, creating a secure shield aesthetic.

6. Conclusion & Verification Summary

The Nexora application has been verified across Web and Mobile formats. The TypeScript codebase is clean, modular, and type-safe. Environment variables, database connection strings, and deep-link schemes are safely configured. This document outlines the complete scope of features developed, providing the Nexora development team with a comprehensive blueprint of the application state as of July 2, 2026.